Solution

ELECTRICITY -02

Class 10 - Science

1.
   (c) P has maximum current.
   Explanation: Current at P is maximum because it is connected in series with battery and hence total current drawn from battery flows through it.
2. **(a)** 583 °C
   Explanation: 583 °C
3.
   (c) 25
   Explanation:
   When a piece of wire of resistance R is cut into five equal parts, the resistance of each part will be $\frac{R}{5}$
   When these parts are connected in parallel, the equivalent resistance of the combination will be given by **:** $\frac{1}{R'} = \frac{5}{R} + \frac{5}{R} + \frac{5}{R} + \frac{5}{R} + \frac{5}{R}$
   Therefore, the ratio R/R' will be 25 **:** 1
4.
   (d) nature of the material
   Explanation: Electrical resistivity of a given metallic wire depends on the number density of free electrons in the conductor which is the nature of the material, i.e., the number of free electrons per unit volume. It also depends on the temperature of the conductor.
5.
   (b) Temperature
   Explanation: Temperature
6.
   (c) one-fourth
   Explanation: When the potential difference between the ends of a fixed resistor is halved, the electric power becomes one-fourth of the actual value.
7.
   (c) 2 times
   Explanation: 2 times
8. **(a)** Half
   Explanation: Half
9.
   (b) Insulator
   Explanation: Insulator
10.
   (b) 5 A
   Explanation: The resistance of a material is given by:
   R = $\rho\frac{l}{A}$
   If l' = 2.5l and R' = 0.5 R
   let A' is the area. New resistance is given by:
   R' = $\rho\frac{l'}{A'}$
   Solving for A'
   A' = $\frac{\rho l'}{R'}$ ...(i)
   Plugging all the values,

A' = $\frac{\rho l'}{R'}$ ...(ii)

Dividing (i) by (ii)

$\frac{A'}{A} = \frac{\frac{\rho(2.5)l}{(0.5R)}}{\frac{\rho l}{R}}$

$\frac{A'}{A}$ = 5

A' = 5A

11.

(d) Zero resistance

Explanation: A superconductor is a material that can conduct electricity with zero resistance, When the temperature of certain material reaches certain critical value called the critical temperature, the resistances of the material completely disappears. Then the material behaves as a superconductor.

12.

(b) 4R

Explanation: The resistance of a uniform metallic conductor is directly proportional to its length (l) and inversely proportional to the area of its cross-section (A). $R = \rho\frac{l}{A}$ where ρ is a constant of proportionality and is called the electrical resistivity of the material of the conductor.

$R_1 = p\frac{l_1}{A_1}$ and $R_2 = \frac{l_2}{A_2}$ and $l_2 = 2l_1$

The volume of the wire remains unchanged.

$\therefore \pi {r_1}^2 l_1 = \pi {r_2}^2 l_2$

$\Rightarrow (\pi {r_1}^2)(l_1) = (\pi {r_2}^2)(l_2)$

$\Rightarrow (A_1)(l_1) = (A_2)(2 \times l_1)$

$\Rightarrow A_2 = \frac{A_1}{2}$

Thus, when the wire is stretched to double its length, the area of cross-section becomes half.

$\therefore R_2 = \rho\frac{l_2}{A_2}$

$\Rightarrow R_2 = \rho\frac{2\times 2\times l_1}{A_1}$

$\Rightarrow R_2 = 4 \times R_1$

Thus, the new resistance becomes four times of the original resistance.

13. **(a)** ohm-m

Explanation: Electrical resistivity (also known as resistivity, specific electrical resistance, or volume resistivity) is an intrinsic property that quantifies how strongly a given material opposes the flow of electric current. A low resistivity indicates a material that readily allows the flow of electric current. Resistivity is commonly represented by the Greek letter ρ (rho). The SI unit of electrical resistivity is the ohm-metre ($\Omega \cdot m$)

14. **(a)** The movement of electrons on their outer most orbital is tightly held together.

Explanation: The movement of electrons on their outer most orbital is tightly held together.

15.

(c) Nature of material

Explanation: Resistivity is 'an intrinsic property of a material' that quantifies how strongly a given material opposes the flow of electric current. The resistivity of a metallic wire depends on the nature of the material. It does not depend on the physical dimensions (length, thickness, shape, etc.) of the metallic wire.

16.

(c) One-fourth

Explanation: We know that

R = $\frac{\rho l}{A}$

Therefore, when the diameter of the wire is doubled, the resistance becomes one-fourth of the actual value.

17.

(d) magnitude of current

Explanation: The heat produced by passing an electric current through a fixed resistor is proportional to the square of the magnitude of the current.

18.

(d) 4A

Explanation: Given,

Potential difference, V = 20V

Resistance, R = 5 Ohms

Current, I = ?

We know that,

V = IR

20 = I $\times$ 5

I = $\frac{20}{5}$ = 4A

19.

(b) 60 Ω

Explanation: The resistance of nichrome is 60 Ω when the resistance of a copper wire is 1 Ω.

20.

(d) 80 Ω

Explanation: We know that,

R = $\frac{\rho l}{A}$

Given that,

l' = 2l

A' = $\frac{A}{2}$

And $\rho' = \rho$ (as the material of the wire is the same)

Therefore,

R' = $\frac{\rho' l'}{A'}$

= $\frac{\rho 2l}{\left(\frac{A}{2}\right)}$

= $\frac{4\rho l}{A}$

= 4R

Therefore, R' = 4 $\times$ 20 = 80Ω

21.

(b) Half

Explanation: When the potential difference is constant and the resistance of a circuit is doubled, the current becomes half.

22.

(c) Metals, Quartz

Explanation: Metals are electropositive in nature and are, therefore, good conductors of electricity. Quartz is a chemical compound consisting of one part silicon and two parts oxygen (SiO_2). It is the most abundant mineral found in Earth surface.

Its unique properties make it one of the most useful natural substances. It has electrical properties and heat resistance that make it valuable in electronic products. Mica has superior electrical properties as an insulator. Rubber is also an insulator.

23.

(d) I, II, III, IV

Explanation:

I. V < E because there will be voltage drop across ammeter also. If voltage across ammeter is V_a then E = V + V_a.

II. Applying Ohm's Law across Voltmeter V = $IR_{voltmeter} \Rightarrow R_{voltmeter} = \frac{V}{I}$

III. As stated in 1st option explanation E = V + V_a

So, potential difference across ammeter is V_a = E - V

IV. Since the cell is ideal, its internal resistance r = 0, hence, by ohm's law, $R_{total} = \frac{E}{I}$

24.

(c) Nature of material

Explanation: The resistivity of the conductor does not depend upon its length or thickness because it is the nature of the material. Resistivity is a material property. It changes with respect to the only temperature.

25.

(d) Four times

Explanation: When the diameter of a resistance wire is halved, the resistance becomes four times.

26.

(c) Becomes 4 times

Explanation: Becomes 4 times

27.

(b) 2A

Explanation: Both the conductors are of same material hence their resistivity is same.

p = $\frac{RA}{I}$

For double the length;

or, $p = \frac{RA'}{2l}$

or, $\frac{RA'}{2l} = \frac{RA}{l}$

or, A = 2A

28. **(a)** 2500°C

Explanation: 2500°C

29. **(a)** the shape of the resistor is changed

Explanation: The resistivity of a material is the inherent property of a material. It is defined as resistance offered by unit length and unit area of cross-section of that material when a known voltage is applied across the ends of the material.

This property of material depends upon only on material and the temperature.

Hence, the shape and size of the material do affect its value.

30.

(d) ammeter is connected in series and the voltmeter in parallel.

Explanation: ammeter is connected in series and the voltmeter in parallel.

31.

(c) 0.67 amp

Explanation: Given,

$R_1 = 0.2\ \Omega$

$R_2 = 0.3\ \Omega$

$R_3 = 0.4\ \Omega$

$R_4 = 0.5\ \Omega$

$R_5 = 12\ \Omega$

V = 9V

Therefore, the resultant resistance is given as:

$R = R_1 + R_2 + R_3 + R_4 + R_5$

= 0.2 + 0.3 + 0.4 + 0.5 + 12

= 13.4 Ω

The current flow through 12 Ω resistance is given as: $\frac{V}{R}$

Therefore, I = $\frac{9}{13.4}$

I = 0.67amp

32. **(a)** 1Ω

Explanation: $R_1 = R_2 = R_3 = 3\Omega$

$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

= $\frac{1}{3} + \frac{1}{3} + \frac{1}{3}$

= $\frac{3}{3}$

= 1

Therefore, R = 1Ω

33.

(d) The reading will be doubled.

Explanation: Let, resistance of bulb P and Q be R. So, reading of ammeter before connecting Q to the circuit is
$I=\frac{V}{R}$ ...(i)
After connecting Q to the circuit, resultant resistance of the circuit is
$\frac{1}{R_{\text{eq}}}=\frac{1}{R}+\frac{1}{R}=\frac{2}{R}$ or $R_{\text{eq}}=\frac{R}{2}\Omega$
Thus, new reading of ammeter will be
$I'=\frac{V}{R/2}=\frac{2V}{R}$
or I' = 2I (From (i))

34.

(d) 1 ohm

Explanation: 1 ohm

35.

(b) Ammeter in series and voltmeter in parallel

Explanation: Ammeter in series and voltmeter in parallel

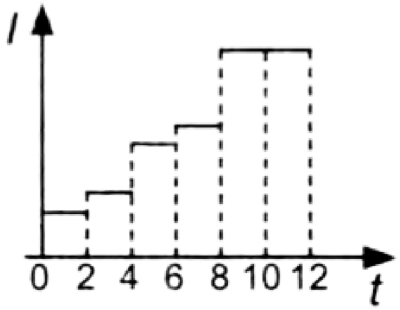
36.

(c) $\frac{R}{4}$

Explanation: Given that the resistance of each part is: $\frac{R}{2}$

Let resultant resistance be R'
$\frac{1}{R'}=\frac{2}{R}+\frac{2}{R}$
R' = $\frac{R}{4}$

37.

(b) 

Explanation: After regular closing of switches, total resistance decreases, thus increasing the current.

38.

(b) 1 Ω

Explanation: In this case, 5 resistors are connected in series and the total resistance is as follows.

Total resistance = $\frac{1}{5}+\frac{1}{5}+\frac{1}{5}+\frac{1}{5}+\frac{1}{5}$ = 1 ohm

Hence, the maximum resistance which can be made using five resistors each of $\frac{1}{5}$ ohm is 1 ohm.

39.

(c) if one lamp fails the others remain lit

Explanation: When the lamps are arranged in parallel connection, the current flowing through them is different. Therefore, if one lamp fails, the others remain lit.

40.

(b) 0.005A

Explanation: The current in a circuit made up of a series of resistors is the same in every area of the circuit or is the same current flowing through every resistor.

41.

(c) 2 A

Explanation: 4 Ω and 2 Ω coils are connected in parallel.

Therefore, the combined resistance, R
$\frac{1}{R}=\frac{1}{4}+\frac{1}{2}=\frac{3}{4}$
R = $\frac{3}{4}\Omega$
Total current, I = $\frac{V}{R}$ = 3 Ω

$\frac{V}{\left(\frac{4}{3}\right)} = 3$

V = 4V

Current through 2 Ω coil = $\frac{V}{2} = \frac{4}{2}$ = 2 A

42. **(a)** 8 ohm

Explanation: R_1 = 2 Ohm

R_2 = 6 Ohm

Let case 1 be a parallel combination

$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$

$\frac{1}{R} = \frac{1}{2} + \frac{1}{6} = \frac{4}{6}$

R = $\frac{6}{4}$ = 1.5 ohm

Let case 2 be series combination

R = R_1 + R_2

= 2 + 6 = 8 ohm

43.

(b) Network (iii)

Explanation:

i. R_1 = kΩ + 200 Ω = 1000 Ω + 200 Ω = 1200 Ω

$\frac{1}{R_i} = \frac{1}{1200} + \frac{1}{1000} = \frac{11}{6000}$

$\therefore R_{\text{i}} = \frac{6000}{11} = 545.45\Omega$

ii. $\frac{1}{R_{\text{ii}}} = \frac{1}{1000} + \frac{1}{200} + \frac{1}{1000} = \frac{7}{1000}$

$\therefore R_{\text{ii}} = \frac{1000}{7} = 142.85\Omega$

iii. $\frac{1}{R_1} = \frac{1}{1000} + \frac{1}{1000} = \frac{2}{1000}$

$R_1 = \frac{1000}{2} = 500\Omega$

R_{iii} = 500 + 200 = 700 Ω

$\therefore$ R_{iii} > R_i > R_{ii}

44. **(a)** 16

Explanation: 16

45. **(a)** same current flowing through them when connected in series

Explanation: In series combination, the current does not change because each resistor receives a common current. In other words, the current does not get divided into branches.

46. **(a)** Gets divided across each component

Explanation: Gets divided across each component

47.

(c) $\frac{r_1+\sqrt{r_1-4r_1r_2}}{r_1-\sqrt{r_1-4r_1r_2}}$

Explanation: r_1 = R_1 + R_2

dividing both side by, R_2, we get

$\frac{r_1}{R_2}$ = $\frac{R_1}{R_2}$ + 1 = X + 1 ...eque. (i)

$\frac{1}{r_2}$ = $\frac{1}{R_1}$ + $\frac{1}{R_2}$

$\frac{R_1}{r_2}$ = 1 + $\frac{R_1}{R_2}$ = 1 + X ... eque. (ii)

(i) $\times$ (ii)

$\left(\frac{r_1}{r_2}\right) \times \left(\frac{R_1}{R_2}\right)$ = $(1 + X)^2$

$\frac{r_1}{r_2}$ X = 1 + X^2 + 2X

$\Rightarrow$ X^2 + $\left(2 - \frac{r_1}{r_2}\right)$ X + 1 = 0

$\frac{R_1}{R_2}$ = X = $\frac{\left(\frac{r_1}{r_2}-2\right)+\sqrt{\left(2-\frac{r_1}{r_2}\right)^2-4}}{2}$

$\frac{r_1+\sqrt{r_1-4r_1r_2}}{r_1-\sqrt{r_1-4r_1r_2}}$

48.

(d) 15.9 V

Explanation: 15.9 V

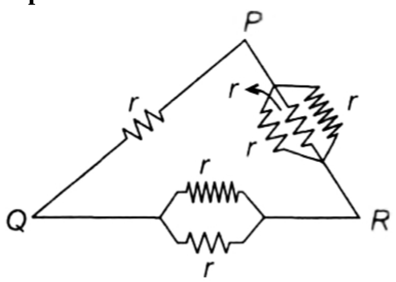
49.

(b) Sum of individual resistance

Explanation: In a series circuit, the current is the same for all of the elements. The total resistance of resistors in series is equal to the sum of their individual resistances.

50. **(a)** P and Q

Explanation: Let resistance of each resistor be r.



Resistance in arm PQ, R_1 = r

Resistance in arm QR, R_2 = $\frac{r\times r}{r+r}=\frac{r}{2}$

Resistance in arm PR,

$R_3=\frac{1}{\frac{1}{r}+\frac{1}{r}+\frac{1}{r}}=\frac{r}{3}$

Let net resistance between P and Q is R_{PQ} and given as follows:

$\frac{1}{R_{PQ}}=\frac{1}{r}+\frac{1}{\frac{r}{3}+\frac{r}{2}}=\frac{1}{r}+\frac{6}{5r}=\frac{11}{5r}$

$\therefore R_{PQ}=\frac{5r}{11}$

Net resistance between Q and R, R_{qr} is given as follows:

$\frac{1}{R_{QR}}=\frac{1}{\frac{r}{2}}+\frac{1}{r+\frac{r}{3}}=\frac{2}{r}+\frac{3}{4r}=\frac{11}{4r}$

$\therefore R_{QR}=\frac{4r}{11}$

Net resistance between P and R, Rpr is given as follows:

$\frac{1}{R_{PR}}=\frac{1}{r/3}+\frac{1}{r+r/2}=\frac{3}{r}+\frac{2}{3r}=\frac{11}{3r}$

Hence R_{PQ} > R_{QR} > R_{PR}

51.

(c)

8Ω

Explanation: Maximum resistance will be in series.

So, 2 + 2 + 2 + 2 = 8 ohm.

52.

(b) 6Ω

Explanation: $R_s = R_1 + R_2 = 4 + 2 = 6\ \Omega$

53.

(c) 55 A

Explanation: 55 A

54.

(d) 21Ω

Explanation: $Rs = R_1 + R_2 + R_3 + R_4 = 1 + 2 + 6 + 12 = 21\ \Omega$

55.

(b) at least two more components in the circuit.

Explanation: Voltmeter, cell and the ammeter are to be connected to find the resistance.

56.

(d) 10Ω

Explanation: Resistance of wires RT and ST are in series, $R_1 = 6 + 6 = 12\Omega$

$\therefore$ Resistance across RS (R_2) is given by

$\frac{1}{R_2} = \frac{1}{12} + \frac{1}{6} = \frac{1}{4}$

$\Rightarrow R_2 = 4\Omega$

Since resistance of wire PR, R_2 and resistance of wire QS are in series.

Resistance between P and Q = 3 + 4 + 3 = 10Ω

57. **(a)** $V_1 = V_2 < V_3$

Explanation: Total potential difference across resistors,

$V = V_1 + V_2 + V_3$

Since the resistors are connected in series, the same current flow through each resistance and

$R_1 = R_2 = R_3 = R_4 = R$

$\therefore V_1 = IR_1 = IR$

$V_2 = IR_2 = IR$

$V_3 = l(R_3 + R_4) = /(R + R) = 2\ IR$

$\therefore V_1 = V_2 < V_3$

58.

(b) 293

Explanation: A total of 293 bulbs can be used when they are all connected in parallel and have 75 W and 220 V.

59. **(a)** 25:1

Explanation: When a piece of wire of resistance R is cut into five equal parts, the resistance of each part will be $\frac{R}{5}$

When these parts are connected in parallel, the equivalent resistance of the combination = $\frac{1}{R'}$ = $\frac{5}{R} \times 5$

Therefore, the ratio R : R' will be 25:1

60.

(d) Current

Explanation: Resistors in series have same current.

61.

(d) 2Ω

Explanation: The resistance of each resistor = $\frac{1}{2}\Omega$

Maximum resistance can be found when the resistors are conned in series combination.

Thus for series combination

$Re = R_1 + R_2 + R_3 + R_4$

$Re = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{4}{2}$

$Re = 2\ \Omega$

62.

(d) It has low resistance

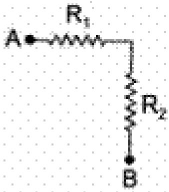
Explanation: An ammeter is used to measure the current flowing through a circuit. The current remains the same in series connection and the resistance of an ammeter is very small due to which it does not affect the current to be measured. So, the proper value of current can be measured. Electric currents are measured in amperes.

63.

(c) S_1, S_3, S_5 are closed

Explanation: Current is maximum when resistance in the circuit is minimum, i.e., when S_1, S_3, S_5 are closed because then all resistances will be short-circuited $I_{\max} = \frac{V}{R}$.

64.

(d) 

Explanation: R_1 and R_2 have one common point not shared by any other device.

65.

(d) 0.06A

Explanation: Both bulb receive the same current because they are connected in series. They will have different voltages.

66. **(a)** 1 Ω

Explanation: Resistors are in series whenever the flow of charge, or the current, must flow through components sequentially. The total resistance in a serially connected circuit is equal to the sum of the individual resistances since the current has to pass through each resistor in sequence through the circuit.

In this case, 5 resistors are connected in series and the total resistance is as follows.

Total resistance $= \frac{1}{5} + \frac{1}{5} + \frac{1}{5} + \frac{1}{5} + \frac{1}{5} = 1$ ohm

Hence, the maximum resistance which can be made using five resistors each of 1/5 ohm is 1 ohm.

67. **(a)** $\frac{1}{25}$ Ω

Explanation: When resistors are connected in parallel, the supply current is equal to the sum of the currents through each resistor. Also, they have the same potential difference across them.

In other words, any components in parallel have the same potential difference across them. And, the total resistance will always be lesser than the least resistance in the circuit.

The total resistance is calculated as $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$. That is, R $= R_T^{-1}$

In this case, the given resistance is $\frac{1}{5}$ ohms each. Therefore, the minimum resistance that can be obtained from the parallel connection of these five resistors is given as follows:

$\frac{1}{R_T} = \frac{5}{1} + \frac{5}{1} + \frac{5}{1} + \frac{5}{1} + \frac{5}{1} = \frac{25}{1}$. Therefore, R = $\frac{1}{25}$

Hence, the minimum resistance which can be made using five resistors each of 1/5 ohms is 1/25 ohms.

68.

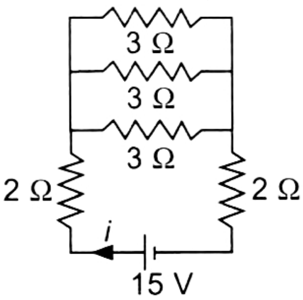
(c) 20 Ω

Explanation: The resultant resistance obtained from the resistors that are placed in series combination is connected with 40 Ω in parallel such that the overall resistance obtained is equal to 20 Ω.

69.

(c) 3 A

Explanation: The given circuit can be redrawn as follows



$\frac{1}{R'} = \frac{1}{3} + \frac{1}{3} + \frac{1}{3}$ or $R' = 1\Omega$

$\therefore$ R_{Total} = 2 + 1 + 2 = 5Ω

Hence, i = $\frac{V}{R} = \frac{15}{5} = 3$ A

70.

(b) 10 V

Explanation: 10 V